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311 CA
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Interferența luminii.
Dispozitivul lui Young.

TABEL DE VALORI

Nr. Crt.	R (cm)	x_1 (mm)	x_2 (mm)	N	i (mm)	$\bar{i}$ (mm)	$\bar{\lambda}$ (nm)	$\sigma_{\bar{i}}$ (mm)
1.	40,1	21,83	24,89	6	0,51	0,509	634	1,22
2.		22,34	25,42		0,513			
3.		23,35	26,42		0,511			
4.		25,93	28,96		0,505			
5.		26,92	29,98		0,51			
6.		27,44	30,48		0,506			
7.		28,46	31,54		0,513			
8.		31,03	34,06		0,505			
9.		32,02	35,09		0,511			
10.		32,54	35,60		0,51			

$d = 0,5 \text{ mm} = 0,5 \cdot 10^{-3} \text{ m}$

PRELUCRAREA DATELOR EXPERIMENTALE

$$i = \frac{x_2 - x_1}{N}$$

$$\rightarrow \bar{i} = \frac{\lambda R}{d} \Rightarrow \lambda = \frac{\bar{i} \cdot d}{R} = \frac{0,509 \cdot 0,5}{401} = 0,000\,634 \text{ mm} = 634 \text{ nm}$$

$$\Rightarrow \lambda = 634 \text{ nm}$$

* Abaterea pătratică:

$$\sigma_{\bar{i}} = \sqrt{\frac{\sum_{k=1}^{n} (i_k - \bar{i})^2}{n(n-1)}}$$

$$\sigma_{\bar{i}} = \sqrt{\frac{3(0,51 - 0,509)^2 + 2(0,513 - 0,509)^2 + 2(0,511 - 0,509)^2 + 2(0,505 - 0,509)^2 + (0,506 - 0,509)^2}{10 \cdot 9}}$$

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$$\sigma_{\bar{i}} = \sqrt{\frac{3\cdot10^{-6} + 2.16\cdot10^{-6} + 8\cdot10^{-6} + 32\cdot10^{-6} + 9\cdot10^{-6}}{9\cdot10}} = \sqrt{\frac{84\cdot10^{-6}}{10\cdot9}} = 0,966\cdot10^{-3}\ mm$$

$$= 9,66\cdot10^{-4}\ m$$

$$\sigma_{\bar{i}}^2 = 0,93\cdot10^{-6}\ mm^2 \Rightarrow \sigma_{\bar{i}}^2 = 0,93\cdot10^{-12}\ m^2$$

$$\sigma_{\bar{R}} = \sqrt{\frac{10\left(\frac{1}{2}\cdot1\ mm\right)^2}{9\cdot10}} = \sqrt{\frac{(0,5)^2\cdot10^{-6}}{9}} = \frac{0,5\cdot10^{-3}}{3} = 0,166\cdot10^{-3}\ m$$

$$\sigma_{\bar{R}}^2 = 0,027\cdot10^{-6}\ m^2$$

$$\sigma_{\bar{\lambda}}^2 = \left(\frac{\partial\lambda}{\partial i}\right)^2\Big|_{\substack{i=\bar{i}\\R=\bar{R}}}\cdot\sigma_{\bar{i}}^2 + \left(\frac{\partial\lambda}{\partial R}\right)^2\Big|_{\substack{i=\bar{i}\\R=\bar{R}}}\cdot\sigma_{\bar{R}}^2$$

$$= \left(\frac{d}{R}\right)^2\cdot\sigma_{\bar{i}}^2 + \left(-\frac{id}{R^2}\right)^2\cdot\sigma_{\bar{R}}^2$$

$$= \left(\frac{0,5\cdot10^{-3}}{40,1\cdot10^{-2}}\right)^2\cdot0,93\cdot10^{-12} + \frac{(0,509)^2\cdot10^{-6}\cdot(0,5)^2\cdot10^{-6}}{(40,1)^4\cdot10^{-8}}\cdot0,027\cdot10^{-6}$$

$$= (0,01246\cdot10^{-1})^2\cdot0,93\cdot10^{-12} + \frac{0,06447\cdot10^{-12}\cdot0,027\cdot10^{-6}}{1608,01\cdot1608,01\cdot10^{-8}}$$

$$= 0,1552\cdot10^{-5}\cdot0,93\cdot10^{-12} + \frac{174,849\cdot10^{-5}\cdot10^{-18}\cdot10^{8}}{1608,01\cdot1608,01}$$

$$= 0,14433\cdot10^{-17} + 0,6763\cdot10^{-19}$$

$$= 1,4433\cdot10^{-18} + 0,06463\cdot10^{-18}$$

$$= 1,5109\cdot10^{-18}$$

$$\Rightarrow \sigma_{\bar{\lambda}} = \sqrt{1,5109\cdot10^{-18}} = 1,22\cdot10^{-9}\ m \Rightarrow \sigma_{\bar{\lambda}} = 1,22\ nm$$

Răspuns: $\lambda \in (\lambda - \sigma_{\bar{\lambda}};\ \lambda + \sigma_{\bar{\lambda}})$

$\lambda \in (634 - 1,22;\ 634 + 1,22) \Rightarrow \lambda \in (632,78;\ 635,22)\ nm$

Hoşu